# Task 1

Assumptions:

1. Customer Number uniquely identifies each customer. Which means, even if two customers have the same name, they still can be distinguished by customer number.
2. There is no special relationship between specific employees and customers.
3. The part name may not be unique, but part number is unique.
4. All parts of the same type are associated with the same cage code. However, two different types of parts may share the same cage code.
5. Each order can be uniquely identified by the combination of customerNumber, orderDate, time, employee and partNumber.
6. One order can include multiple parts, and one part can appear in multiple orders.

The primary key in 1NF should be (CustomerNumber, OrderDate, Time, PartNumber).

- CustomerNumber alone is not unique, since the same customer can place multiple orders.
- OrderDate and Time alone are not unique, as many customers can place orders at the same time.
- PartNumber determines PartName (since each part number corresponds to exactly one part name), but PartNumber alone is not unique in this context because a customer may order the same part multiple times or in different quantities.
- However, the combination of CustomerNumber, OrderDate, Time, and PartNumber uniquely identifies one part ordered by a specific customer.

| **customerNumber** | customerName | customerType | **orderDate** | **time** | employee | **partNumber** | partName | partType | cageCode | quantityOrdered | unitPrice |
|---|---|---|---|---|---|---|---|---|---|---|---|
| HG54587 | Jeff Peterson | Consumer | 7/1/2024 | 10:30 am | D. Harrison | 10654 | Float Control | Plumbing | G413 | 4 | 12 |
| HG54587 | Jeff Peterson | Consumer | 7/1/2024 | 10:30 am | D. Harrison | 10456 | Modulator | Electrical | H433 | 3 | 7 |
| HG54587 | Jeff Peterson | Consumer | 7/1/2024 | 10:30 am | D. Harrison | 10776 | Hose Assembly | Plumbing | G413 | 7 | 9 |
| HG54587 | Jeff Peterson | Consumer | 7/1/2024 | 10:30 am | D. Harrison | 10657 | Float Assembly | Plumbing | G413 | 5 | 10 |

This database is in 1NF since each of its columns contains only atomic values, and there are no repeating groups of columns or rows.

However, there is a lot of redundancy in this version: For instance, CustomerName repeats with CustomerNumber, and there are duplicate Employee entries. Besides, there exist partial and transitive dependencies that require further normalization.

Partial Dependency needs to be resolved in 2NF, which happens when a non-key attribute depends on only part of a composite key.

part

| **partNumber** | partName | partType | cageCode | unitPrice |
|---|---|---|---|---|
| 10654 | Float Control | Plumbing | G413 | 12 |
| 10456 | Modulator | Electrical | H433 | 7 |
| 10776 | Hose Assembly | Plumbing | G413 | 9 |
| 10657 | Float Assembly | Plumbing | G413 | 10 |

PK: partNumber

order

PK: (customerNumber, orderDate, time, partNumber)

| **customerNumber** | **orderDate** | **time** | **partNumber** | quantityOrdered |
|---|---|---|---|---|
| HG54587 | 7/1/2024 | 10:30 am | 10654 | 4 |
| HG54587 | 7/1/2024 | 10:30 am | 10456 | 3 |
| HG54587 | 7/1/2024 | 10:30 am | 10776 | 7 |
| HG54587 | 7/1/2024 | 10:30 am | 10657 | 5 |

customer

| **customerNumber** | customerName | customerType |
|---|---|---|
| HG54587 | Jeff Peterson | Consumer |

PK: customerNumber

employee

| **customerNumber** | **orderDate** | **time** | employee |
|---|---|---|---|
| HG54587 | 7/1/2024 | 10:30 am | D. Harrison |

PK: (customerNumber, orderDate, time)

This database is now in 2NF since it is in 1NF and all the non-PK fields are dependent on All the PK fields.

What I split:

1. From (CustomerNumber, OrderDate, Time, PartNumber) → Employee, QuantityOrdered —— make Order .
2. From CustomerNumber → CustomerName, CustomerType —— make Customer.
3. From PartNumber → PartName, PartType, CageCode, UnitPrice —— make Part.

In 2NF, every non-key attribute depends on the whole key, but some attributes still depend on other non-key attributes. For instance, CageCode did not depend on PartNumber directly, it depended on PartType, whichi is a transitive dependency.

cage_code

| **partType** | cageCode |
|---|---|
| Plumbing | G413 |
| Electrical | H433 |

Foreign Key FK here used to establish a one-to-many relationship, which is partType.

customer

| **customerNumber** | customerName | customerType |
|---|---|---|
| HG54587 | Jeff Peterson | Consumer |

part

| **partNumber** | partName | *partType* | unitPrice |
|---|---|---|---|
| 10654 | Float Control | Plumbing | 12 |
| 10456 | Modulator | Electrical | 7 |
| 10776 | Hose Assembly | Plumbing | 9 |
| 10657 | Float Assembly | Plumbing | 10 |

order

| ***customerNumber*** | **orderDate** | **time** | ***partNumber*** | quantityOrdered |
|---|---|---|---|---|
| HG54587 | 7/1/2024 | 10:30 am | 10654 | 4 |
| HG54587 | 7/1/2024 | 10:30 am | 10456 | 3 |
| HG54587 | 7/1/2024 | 10:30 am | 10776 | 7 |
| HG54587 | 7/1/2024 | 10:30 am | 10657 | 5 |

employee

| **customerNumber** | **orderDate** | **time** | employee |
|---|---|---|---|
| HG54587 | 7/1/2024 | 10:30 am | D. Harrison |

Now this database is in 3NF.
Since it already 2NF and no non-key attribute depends on another non-key attribute (transitive dependency).

# Task 2

Assumptions:
1. Each therapist has a unique staffNo and work at one or more branches.
2. Each patient has a unique patNo.
3. branchNo uniquely identifies a branch.
4. Each appointment is for a specific date and time at a branch with a therapist.

| staffNo | therapistName | patNo | patName | appointmentDateTime | branchNo |
|---|---|---|---|---|---|
| S1011 | Fred Smith | P100 | Lily White | 9/12/2022 10:00 | M15 |
| S1011 | Fred Smith | P105 | Jill Baker | 9/12/2022 12:00 | M15 |
| S1024 | Heidi Pierce | P108 | Andy McKee | 9/12/2022 10:00 | Q10 |
| S1024 | Heidi Pierce | P108 | Andy Mckee | 9/14/2022 14:00 | Q10 |
| S1032 | Richard Levin | P105 | Jill Baker | 9/14/2022 16:30 | M15 |
| S1032 | Richard Levin | P110 | Jimmy Winter | 9/15/2022 18:00 | B13 |

This original table is not in 1NF yet, since in appointmentDateTime, multiple values in one field of a record.

| **staffNo** | therapistName | **patNo** | patName | **date** | **time** | branchNo |
|---|---|---|---|---|---|---|
| S1011 | Fred Smith | P100 | Lily White | 9/12/2022 | 10:00 | M15 |
| S1011 | Fred Smith | P105 | Jill Baker | 9/12/2022 | 12:00 | M15 |
| S1024 | Heidi Pierce | P108 | Andy McKee | 9/12/2022 | 10:00 | Q10 |
| S1024 | Heidi Pierce | P108 | Andy Mckee | 9/14/2022 | 14:00 | Q10 |
| S1032 | Richard Levin | P105 | Jill Baker | 9/14/2022 | 16:30 | M15 |
| S1032 | Richard Levin | P110 | Jimmy Winter | 9/15/2022 | 18:00 | B13 |

After splitting appointmentDateTime into date and time, this database is now in 1NF since each of its columns contains only atomic values, and there are no repeating groups of columns or rows.

The primary key in 1NF should be (staffNo, patNo, date, time), which uniquely represents each appointment.

However, redundancy exists:
1. therapistName repeats for the same staffNo. staffNo → therapistName
2. patName repeats for the same patNo. patNo → patName
3. branchNo is depends on staffNo and date since the therapist only see patients at one specific branch on any given day.

staff

| **staffNo** | therapistName |
|---|---|
| S1011 | Fred Smith |
| S1024 | Heidi Pierce |
| S1032 | Richard Levin |

patient

| **patNo** | patName |
|---|---|
| P100 | Lily White |
| P105 | Jill Baker |
| P108 | Andy McKee |
| P110 | Jimmy Winter |

appointment

| *staffNo* | ***patNo*** | ***date*** | **time** |
|---|---|---|---|
| S1011 | P100 | 9/12/2022 | 10:00 |
| S1011 | P105 | 9/12/2022 | 12:00 |
| S1024 | P108 | 9/12/2022 | 10:00 |
| S1024 | P108 | 9/14/2022 | 14:00 |
| S1032 | P105 | 9/14/2022 | 16:30 |
| S1032 | P110 | 9/15/2022 | 18:00 |

branch

| **staffNo** | branchNo | **date** |
|---|---|---|
| S1011 | M15 | 9/12/2022 |
| S1024 | Q10 | 9/12/2022 |
| S1024 | Q10 | 9/14/2022 |
| S1032 | M15 | 9/14/2022 |
| S1032 | B13 | 9/15/2022 |

The table is currently in 3NF. I have divided it into four parts: staff, patient, appointment, and branch. The primary key (PK) for staff is staffNo, for patient it is patNo, for appointment it is (patNo, date, time), and for branch it is (staffNo, date). Here, patNo and staffNo also serve as foreign keys (FK).

# Task 3

Assumptions:
1. Each eNo uniquely identifies an employee, and each employee has exactly one eName.
2. Each contractNo applies to exactly one eventNo.
3. Each eventNo corresponds to exactly one eventLoc.
4. hours represents the number of hours an employee worked on a specific contract.
5. An employee may work on multiple contracts for the same event.

| **eNo** | **contractNo** | hours | eName | eventNo | eventLoc |
|---|---|---|---|---|---|
| 1135 | C1024 | 16 | Smith J | H25 | Queens |
| 1057 | C1024 | 24 | Hocine D | H25 | Queens |
| 1068 | C1025 | 28 | White T | H4 | Yonkers |
| 1135 | C1025 | 15 | Smith J | H4 | Yonkers |
| 1135 | C1026 | 10 | Smith J | H25 | Queens |

This database is in 1NF since each of its columns contains only atomic values, and there are no repeating groups of columns or rows.

Primary Key for this database is (eNo, contractNo), which uniquely identify each record.

However, partial dependency needs to be resolved in 2NF, which happens when a non-key attribute depends on only part of a composite key. For instance, eName depends only on eNo,. Similarly, eventNo and eventLoc depend only on contractNo.

employee

| **eNo** | eName |
|---|---|
| 1135 | Smith J |
| 1057 | Hocine D |
| 1068 | White T |

event

| eventNo | **contractNo** | eventLoc |
|---|---|---|
| H25 | C1024 | Queens |
| H4 | C1025 | Yonkers |
| H25 | C1026 | Queens |

working_hours

| **eNo** | **contractNo** | hours |
|---|---|---|
| 1135 | C1024 | 16 |
| 1057 | C1024 | 24 |
| 1068 | C1025 | 28 |
| 1135 | C1025 | 15 |
| 1135 | C1026 | 10 |

This database is currently in 2NF because all partial dependencies have been resolved. Each non-key attribute now fully depends on the whole primary key of its relation.

However, the database is not yet in 3NF, because there is still a transitive dependency. Specifically, eventLoc does not directly depend on the primary key of the Contract relation (contractNo), but rather indirectly through eventNo.

To achieve 3NF, we need to further decompose the tables to eliminate this transitive dependency.

After this split, all non-key attributes will depend only on their relation's primary key, satisfying 3NF.

employee

| **eNo** | eName |
|---|---|
| 1135 | Smith J |
| 1057 | Hocine D |
| 1068 | White T |

contract

| *eventNo* | **contractNo** |
|---|---|
| H25 | C1024 |
| H4 | C1025 |
| H25 | C1026 |

working_hours

| ***eNo*** | ***contractNo*** | hours |
|---|---|---|
| 1135 | C1024 | 16 |
| 1057 | C1024 | 24 |
| 1068 | C1025 | 28 |
| 1135 | C1025 | 15 |
| 1135 | C1026 | 10 |

event

| **eventNo** | eventLoc |
|---|---|
| H25 | Queens |
| H4 | Yonkers |

In the Contract table, eventNo now serves as a foreign key FK referencing the Event table.